Week 2 - Linear Regression

Essential Study Guide

1. What is the lecture about?

Regression predicts a **numerical outcome**. We observe data

$$(x_1, y_1), \ldots, (x_N, y_N),$$

choose a family of prediction functions f_θ, measure their errors with a loss, and choose the parameters θ that minimise the empirical loss.

Core workflow: choose a model family → make predictions → compute residuals → compute loss → adjust parameters to reduce the loss.

Main notation

Symbol	Meaning
N	Number of observations in the dataset.
i	Observation index, $i = 1, \ldots, N$.
x_i	Input of observation i (for one input feature).
y_i	Observed target/output of observation i.
f_{true}	Unknown true relationship between input and output.
ε_i	Noise for observation i.
f_θ	Predictor controlled by parameters θ.
$\hat{y}_i = f_\theta(x_i)$	Model prediction for observation i.
$r_i(\theta) = f_\theta(x_i) - y_i$	Residual = prediction minus target.
ℓ	Loss for **one** prediction-target pair.
$\mathcal{L}$	Empirical loss over the **whole dataset**.
θ^*	A parameter choice that minimises the empirical loss.

Underlying modelling idea

The observed target is represented as

$$y_i = f_{\text{true}}(x_i) + \varepsilon_i.$$

We observe x_i and y_i, but not f_{true} or ε_i. A simple affine model is

$$f_\theta(x) = b + ax,$$

where a is the slope and b is the intercept (bias).

During fitting, the data x_i, y_i are fixed. The parameters such as a, b or θ are the quantities that vary.

2. Residuals, loss and the fitting objective

Residual

For one observation,

$$r_i(\theta) = \hat{y}_i - y_i = f_\theta(x_i) - y_i.$$

A positive residual means overprediction; a negative residual means underprediction.

Two common losses

$$\ell_{\text{abs}}(\hat{y}, y) = |\hat{y} - y|, \qquad \ell_{\text{sq}}(\hat{y}, y) = (\hat{y} - y)^2.$$

Squared loss is used throughout the main derivations because it is differentiable everywhere and penalises large errors more strongly.

Empirical squared loss

For parameters θ,

$$\boxed{\mathcal{L}_{\text{sq}}(\theta) = \sum_{i=1}^{N} \big(f_\theta(x_i) - y_i\big)^2 = \sum_{i=1}^{N} r_i(\theta)^2.}$$

The fitting problem is

$$\boxed{\theta^* \in \arg\min_{\theta} \mathcal{L}_{\text{sq}}(\theta).}$$

Do not confuse the two levels: ℓ_{sq} scores one prediction; $\mathcal{L}_{\text{sq}}$ adds the losses for all observations.

Why derivatives appear

At a differentiable local minimum, Fermat's theorem gives

$$\nabla \mathcal{L}_{\text{sq}}(\theta^*) = 0.$$

So the usual strategy is:

1. differentiate the empirical loss;
2. set the derivative/gradient to zero;
3. solve for stationary points;
4. check that the stationary point is actually a minimum.

What the fitted loss does - and does not - tell us

A small empirical loss means the model fits the **observed training data** well. It does not by itself guarantee accurate predictions on unseen data. Performance on unseen data is called **generalisation**.

Multiplying the empirical loss by a positive constant, such as $1/N$ or $1/(2N)$, does not change the minimiser. It only rescales the loss and its gradient.

3. Three one-input predictor families

A. Constant predictor

$$f_b(x) = b, \qquad \mathcal{L}_{\text{sq}}^{\text{const}}(b) = \sum_{i=1}^{N}(b - y_i)^2.$$

Setting the derivative to zero gives

$$\boxed{b^* = \bar{y} = \frac{1}{N}\sum_{i=1}^{N} y_i.}$$

So under squared loss, the best constant prediction is the **mean of the targets**.

B. Line through the origin

$$f_a(x) = ax, \qquad \mathcal{L}_{\text{sq}}^{\text{slope}}(a) = \sum_{i=1}^{N}(ax_i - y_i)^2.$$

If $\sum_i x_i^2 > 0$,

$$\boxed{a^* = \frac{\sum_{i=1}^{N} x_i y_i}{\sum_{i=1}^{N} x_i^2}.}$$

This model is forced to pass through $(0, 0)$.

C. General affine line

$$f_\theta(x) = b + ax, \qquad \theta = \begin{pmatrix} b \\ a \end{pmatrix},$$

$$\mathcal{L}_{\text{sq}}^{\text{aff}}(b, a) = \sum_{i=1}^{N}(b + ax_i - y_i)^2.$$

Define

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i, \qquad \bar{y} = \frac{1}{N}\sum_{i=1}^{N} y_i.$$

If the inputs are not all equal, the unique minimiser is

$$\boxed{a^* = \frac{\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{N}(x_i - \bar{x})^2}, \qquad b^* = \bar{y} - a^*\bar{x}.}$$

The fitted line passes through $(\bar{x}, \bar{y})$.

Lecture example: for $x = (-2, -1, 0, 1, 2)$ and $y = (-1.5, 0.5, 1, 2.5, 7.5)$, $\bar{x} = 0$, $\bar{y} = 2$, so $a^* = 2$, $b^* = 2$, and $f_{\theta^*}(x) = 2 + 2x$. Its squared loss is 6.

How the model family changes flexibility

Family	Prediction	Restriction
Constant	b	Cannot change with x.
Through origin	ax	Must predict 0 at $x = 0$.
Affine	$b + ax$	Can shift and change slope, but cannot curve.

4. Several input features and matrix least squares

Suppose each observation has d features,

$$\mathbf{x} = (x_1, \ldots, x_d) \in \mathbb{R}^{1\times d}.$$

Give each feature a weight and include an intercept:

$$\theta = \begin{pmatrix} b \\ w_1 \\ \vdots \\ w_d \end{pmatrix} \in \mathbb{R}^{(d+1)\times 1}, \qquad f_\theta(\mathbf{x}) = b + \mathbf{x}\mathbf{w}.$$

Design matrix

For N observations,

$$X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1d} \\ \vdots & \vdots & & \vdots \\ 1 & x_{N1} & \cdots & x_{Nd} \end{pmatrix} \in \mathbb{R}^{N\times(d+1)}, \qquad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_N \end{pmatrix}.$$

Each row is one observation. The first column of ones handles the intercept.

Predictions, residuals and loss

$$\boxed{\hat{\mathbf{y}} = X\theta}$$

$$\boxed{\mathbf{r}(\theta) = X\theta - \mathbf{y}}$$

$$\boxed{\mathcal{L}_{\text{sq}}(\theta) = \|X\theta - \mathbf{y}\|_2^2.}$$

Gradient and normal equations

$$\boxed{\nabla\mathcal{L}_{\text{sq}}(\theta) = 2X^\top(X\theta - \mathbf{y}).}$$

Setting the gradient to zero gives the **normal equations**:

$$\boxed{X^\top X\,\theta = X^\top \mathbf{y}.}$$

If $X^\top X$ is invertible,

$$\boxed{\theta^* = (X^\top X)^{-1}X^\top \mathbf{y}.}$$

This is the unique global minimiser.

Dimension check: $X : N \times (d+1)$, $\theta : (d+1) \times 1$, $X\theta : N \times 1$, $X^\top X : (d+1) \times (d+1)$.

The matrix in the normal equations is $X^\top X$, not $XX^\top$. If $X^\top X$ is not invertible, the inverse formula cannot be used and the coefficients need not be unique.

5. Nonlinear features - still linear regression

A straight line may be too restrictive. We can transform the input using fixed feature functions.

Feature map

Choose

$$\phi_1, \dots, \phi_m : \mathbb{R} \to \mathbb{R}$$

and define

$$\phi(x) = \big(\phi_1(x), \dots, \phi_m(x)\big), \qquad \theta = \begin{pmatrix} \theta_1 \\ \vdots \\ \theta_m \end{pmatrix}.$$

Then

$$\boxed{f_\theta(x) = \phi(x)\theta = \sum_{j=1}^{m} \theta_j \phi_j(x).}$$

Quadratic example

Choose

$$\phi(x) = (1, x, x^2), \qquad \theta = \begin{pmatrix} b \\ w_1 \\ w_2 \end{pmatrix}.$$

Then

$$f_\theta(x) = b + w_1 x + w_2 x^2.$$

The model is nonlinear in the input x, but it is **linear in the fitted coefficients** b, w_1, w_2. Therefore it is still a linear regression model in the sense used in this lecture.

Transformed design matrix

$$\Phi = \begin{pmatrix} \phi(x_1) \\ \vdots \\ \phi(x_N) \end{pmatrix} \in \mathbb{R}^{N \times m}.$$

Everything from ordinary least squares carries over by replacing X with Φ:

$$\mathcal{L}^{\phi}_{\text{sq}}(\theta) = \|\Phi\theta - \mathbf{y}\|_2^2,$$

$$\boxed{\Phi^\top \Phi\, \theta = \Phi^\top \mathbf{y},}$$

$$\boxed{\theta^* = (\Phi^\top \Phi)^{-1} \Phi^\top \mathbf{y}} \qquad \text{if } \Phi^\top \Phi \text{ is invertible.}$$

Main idea: nonlinear features change what shapes the model can represent, but the fitting machinery stays the same as long as the predictor is linear in the adjustable coefficients.

What is not linear in the parameters?

A model such as

$$f_w(x) = \sin(wx)$$

is not linear in its adjustable parameter w, so it cannot be handled by the same fixed-feature least-squares form.

6. Gradient descent

The normal equations can give the minimiser directly, but for many parameters it may be preferable to approach the minimum iteratively.

General update

Starting from $\theta^{(0)}$,

$$\boxed{\theta^{(t+1)} = \theta^{(t)} - \eta \nabla \mathcal{L}\big(\theta^{(t)}\big),}$$

where $\eta > 0$ is the learning rate and t is the iteration number.

A small move in the negative-gradient direction tends to reduce the objective because

$$\mathcal{L}(\theta + \Delta) \approx \mathcal{L}(\theta) + \nabla \mathcal{L}(\theta)^\top \Delta,$$

and with $\Delta = -\eta \nabla \mathcal{L}(\theta)$ the change is approximately

$$-\eta \|\nabla \mathcal{L}(\theta)\|_2^2 < 0$$

when the gradient is nonzero.

Linear regression update in matrix form

Since

$$\nabla \mathcal{L}_{\text{sq}}(\theta) = 2X^\top (X\theta - \mathbf{y}),$$

the update becomes

$$\boxed{\theta^{(t+1)} = \theta^{(t)} - 2\eta X^\top \big(X\theta^{(t)} - \mathbf{y}\big).}$$

Equivalently:

$$\mathbf{r}^{(t)} = X\theta^{(t)} - \mathbf{y}, \qquad \theta^{(t+1)} = \theta^{(t)} - 2\eta X^\top \mathbf{r}^{(t)}.$$

For a straight line $b + ax$

$$r_i^{(t)} = b^{(t)} + a^{(t)} x_i - y_i,$$

$$\boxed{b^{(t+1)} = b^{(t)} - 2\eta \sum_i r_i^{(t)},}$$

$$\boxed{a^{(t+1)} = a^{(t)} - 2\eta \sum_i x_i r_i^{(t)}.}$$

Both updates must use the **same current residuals** before either parameter is changed.

Lecture example: starting from $(b^{(0)}, a^{(0)}) = (0, 0)$ with $\eta = 0.04$, the first update gives $(b^{(1)}, a^{(1)}) = (0.8, 1.6)$. The loss falls from 66 to 14.8, while the exact optimum is $(2, 2)$ with loss 6.

The learning rate matters. A step that is too large can increase the loss or make the iterates diverge.

Final checklist

- Regression predicts numbers.
- Residual = prediction − target.
- Least squares minimises the sum of squared residuals.
- For several features: $\hat{\mathbf{y}} = X\theta$ and $\mathcal{L}_{\text{sq}} = \|X\theta - \mathbf{y}\|_2^2$.
- Closed-form solution (when invertible): $\theta^* = (X^\top X)^{-1} X^\top \mathbf{y}$.
- Gradient descent repeatedly moves parameters against the gradient.
- Fixed nonlinear features can create curved predictors while keeping the same least-squares machinery.

Condensed from the Week 2 Linear Regression lecture notes. Proof details, extended examples, exercises and the appendix were intentionally omitted.